

Sampling, Padé, Mellin, Resolvent, and Product–Integral Representations of the Fabius–Rvachev–Thue–Morse System

A comprehensive audited atlas with second-wave frontier results

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Abstract

This report gives a comprehensive, snapshot-relative audit of the recursive LaTeX corpus under `Analysis/FabiusFunction/docs`. The audit covers the 67-path pinned ledger at commit `32d6d36c51d803289e6d6a0dc0c37753766eba47`, the living primary and frontier manuscripts, the Thue–Morse formula atlases, the Fourier-decay campaign, the inverse-function and self-sampling drafts, the archive lineages, and the paper-source transcriptions. Existing results are organized into a representation atlas spanning probability laws, infinite convolutions, finite Thue–Morse splines, sinc and cosine products, Fourier inversion, gamma and zero products, moment/cumulant/Bell algebra, q-binomial formulas, exact dyadic arithmetic, Lambert– W endpoint asymptotics, Cauchy transforms, one-sided transforms, Fredholm determinants, and Jacobi continued fractions. Repository results, immediate recombinations, and new corpus-relative results are explicitly distinguished.

The first-wave frontier constructions are retained: the Cauchy–Stieltjes transform of Rvachev’s up-function, its logarithmic dyadic fixed point, the functional-differential law

$$R'(z) = 2\{R(2z + 1) - R(2z - 1)\},$$

the all-orders Thue–Morse derivative tree, finite logarithmic resolvents, exact transforms of F , one-sided up, and F^{-1} , the cotangent/digamma/Hurwitz-zeta hierarchy for Φ'/Φ , a trace-class Fredholm determinant for Φ , and the positive Jacobi fraction of the up-law.

The second wave develops several different representation mechanisms. Compact support and the vanishing integer samples of Φ give the sparse half-lattice reconstruction

$$\Phi(z) = \operatorname{sinc}_\pi(2z) + \sum_{k \geq 0} \Phi(k + \tfrac{1}{2}) [\operatorname{sinc}_\pi(2z - 2k - 1) + \operatorname{sinc}_\pi(2z + 2k + 1)],$$

and hence a convergent Mittag–Leffler expansion whose residues recover both signed and unsigned Thue–Morse. Fourier duality gives odd-harmonic cosine series for up and F and the infinite weighted Prouhet identities

$$\sum_{k \geq 0} (-1)^{s_2(k)} |\Phi(k + \tfrac{1}{2})| (2k + 1)^{2r} = \begin{cases} \frac{1}{2}, & r = 0, \\ 0, & r \geq 1. \end{cases}$$

A Bose–Malmsten kernel converts the sinc product into an exponential integral and yields all logarithmic derivatives. Mellin transforms of up and F , the quantile moments of F^{-1} , and the generalized binomial series of the up-moments are unified by one entire function.

For the signed Thue–Morse generating function $E(z)$, a new formal orthogonality analysis proves

$$p_{2n}(x) = p_n(x^2), \quad a_n = (-1)^{n+1},$$

and gives an exact nonlinear dyadic recursion for the remaining Jacobi coefficients. At $N = 2^m$, $p_N(x) = x^N + 1$, producing the closed Padé approximant

$$E(z) = \frac{\prod_{j < m} (1 - z^{2^j})}{1 + z^N} + O(z^{2N}).$$

Its roots-of-minus-one partial fractions furnish an N -node complex quadrature that reproduces $2N$ consecutive Thue–Morse signs. Krawtchouk, contour, residue, integral, and antiperiodic Fourier representations of the signed and unsigned words are collected as well. A fully commented Python program, exact rational data, high-precision residuals, and four reproducible figures accompany the report. Every theorem labelled as new is proved here; conjectures are clearly separated from established statements.

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1 Executive summary and novelty boundary

1.1 What was audited

The repository contains several kinds of LaTeX source that must not be conflated:

1. a living primary synthesis of the formalized and analytic theory;
2. a large consolidated non-formalized frontier volume;
3. current draft reports on arithmetic rays, half-integer spectra, self-sampling, inverse asymptotics, Newton/exponent-sequence hierarchies, Fourier decay, and related topics;
4. archived earlier lineages, retained because some contain historically distinctive formulas even when superseded;
5. seven paper-source transcriptions or corrected transcriptions.

The repository’s own pinned audit ledger records 67 LaTeX paths at the following commit:

32d6d36c51d803289e6d6a0dc0c37753766eba47.

The live tree inspected on 27 August 2026 contains additional current drafts, notably the self-sampling, inverse, and Newton/Rvachev reports. Appendix B records the pinned ledger and the live additions used in this study.

1.2 Non-duplication protocol

Every formula in the atlas is tagged conceptually as follows.

Tag	meaning
K	already explicit in the audited repository or classical source; included for orientation
D	an immediate transform or recombination of known identities; useful but not claimed as new
N	apparently absent from the audited corpus; proved or conjectured here

The word “new” throughout this report means *new relative to the audited repository*. No claim of global bibliographic priority is made without a separate literature search. In particular, the current frontier report explicitly proposes a Jacobi continued fraction for the moment generating function or Cauchy transform as an open direction. The present report constructs that fraction and develops its consequences.

1.3 Main new results

The most substantial additions are:

1. the Cauchy transform R and its moment, Laplace-product, Fourier–Laplace, boundary-value, and logarithmic fixed-point representations;
2. the exact resolvent equation and its all-orders Thue–Morse derivative tree;
3. a finite-level 2^N -term logarithmic formula converging to R , paired by boundary jumps with the finite Thue–Morse spline density;

4. the Jacobi J -fraction of R , the contracted Stieltjes S -fraction of the even moments, exact rational coefficients through order 40, Gaussian quadrature convergents, and the theorem $\beta_n \rightarrow 1/4$;
5. a Fredholm determinant realization of the Fourier image and a trace-resolvent interpretation of its logarithmic derivative;
6. cotangent, tangent, valuation-weighted partial-fraction, digamma, and Hurwitz-zeta representations of the logarithmic derivatives of the Fourier image;
7. exact one-sided Laplace and Fourier transforms of up , exact transforms of F on $[0, 1]$, and a master pushforward identity for F^{-1} .

2 Normalization and structural dictionary

2.1 The functions

Let $F : [0, 1] \rightarrow [0, 1]$ denote the bounded Fabius function [4, 2], extended by 0 on $(-\infty, 0]$ and by 1 on $[1, \infty)$. On the unit interval it satisfies

$$F(0) = 0, \quad F(1) = 1, \quad F(1 - x) = 1 - F(x), \quad (1)$$

$$F'(x) = 2F(2x) \quad (0 \leq x \leq \tfrac{1}{2}). \quad (2)$$

The Rvachev function is written

$$\rho(x) := \text{up}(x) = \begin{cases} F(x+1), & -1 \leq x \leq 0, \\ F(1-x), & 0 \leq x \leq 1, \\ 0, & |x| \geq 1. \end{cases} \quad (3)$$

Then ρ is even, strictly positive on $(-1, 1)$, normalized by $\int \rho = 1$, flat at ± 1 , and satisfies

$$\rho'(x) = 2(\rho(2x+1) - \rho(2x-1)). \quad (4)$$

2.2 Fourier convention and basic transforms

We use

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i \xi x} \, dx, \quad f(x) = \int_{\mathbb{R}} \widehat{f}(\xi) e^{2\pi i \xi x} \, d\xi. \quad (5)$$

Set

$$\Phi(\xi) := \widehat{\rho}(\xi), \quad \text{sinc}_{\pi}(z) := \frac{\sin(\pi z)}{\pi z}, \quad \text{sinc}_{\pi}(0) = 1. \quad (6)$$

The classical products are

$$\Phi(z) = \prod_{n=0}^{\infty} \text{sinc}_{\pi}\left(\frac{z}{2^n}\right) = \prod_{m=1}^{\infty} \cos\left(\frac{\pi z}{2^m}\right)^m. \quad (7)$$

The scaling law is

$$\Phi(z) = \text{sinc}_{\pi}(z) \Phi(z/2). \quad (8)$$

2.3 The probability model

Let $U_1, U_2, \dots$ be independent random variables, uniform on $[-1, 1]$, and set

$$X = \sum_{k=1}^{\infty} 2^{-k} U_k. \quad (9)$$

Then X has density ρ . Equivalently,

$$X \stackrel{d}{=} \frac{U + X'}{2}, \quad (10)$$

where $U \sim \text{Unif}[-1, 1]$ and X' is an independent copy of X . The moment generating function is

$$M(t) := \mathbb{E}e^{tX} = \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k} = \Phi\left(\frac{it}{2\pi}\right). \quad (11)$$

The even moments are

$$\mu_{2n} := \mathbb{E}X^{2n} = \int_{-1}^1 x^{2n} \rho(x) \, dx, \quad \mu_{2n+1} = 0. \quad (12)$$

Finally, if

$$Y = \frac{X + 1}{2}, \quad (13)$$

then F is the distribution function of Y , and

$$F'(x) = 2\rho(2x - 1), \quad 0 < x < 1. \quad (14)$$

3 Audited representation atlas

This section compresses the principal representation families already present in the repository and places the new formulas beside them. Proofs of the new entries begin in Section 4.

3.1 Real-space and probability representations of ρ

Infinite convolution (K). Since $2^{-k}U_k$ is uniform on $[-2^{-k}, 2^{-k}]$,

$$\rho = \bigast_{k=1}^{\infty} \left(2^{k-1} \mathbf{1}_{[-2^{-k}, 2^{-k}]} \right). \quad (15)$$

This is the real-space counterpart of the sinc product.

Infinite cube integral (D). For every bounded continuous test function g ,

$$\int_{-1}^1 g(x) \rho(x) \, dx = \int_{[-1, 1]^{\mathbb{N}}} g\left(\sum_{k \geq 1} 2^{-k} u_k\right) \prod_{k \geq 1} \frac{du_k}{2}. \quad (16)$$

Formally, this may be written as a Dirac-delta integral for $\rho(x)$.

Finite Thue–Morse splines (K). Put

$$X_N = \sum_{k=1}^N 2^{-k} U_k, \quad a_{N,j} = \frac{2^N - 1 - 2j}{2^N}, \quad \varepsilon_j = (-1)^{\text{wt}_2(j)}. \quad (17)$$

The density ρ_N of X_N is

$$\rho_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \varepsilon_j (x + a_{N,j})_+^{N-1}. \quad (18)$$

For each fixed derivative order, $\rho_N \rightarrow \rho$ uniformly after N is sufficiently large relative to that order. The repository develops several centered and one-sided variants with explicit error control.

Fourier inversion and cosine form (K). Because Φ is real and even,

$$\rho(x) = \int_{\mathbb{R}} \Phi(\xi) e^{2\pi i x \xi} d\xi, \quad (19)$$

$$\rho(x) = 2 \int_0^\infty \Phi(\xi) \cos(2\pi x \xi) d\xi. \quad (20)$$

The audited corpus also contains Poisson-summation formulas, Legendre expansions, periodized cosine series, dyadic spline limits, and endpoint transseries. They are not rederived here.

3.2 Product, zero, gamma, and exponential representations of Φ

Canonical zero product (K). Let $\nu_2(n)$ be the exponent of 2 in n . The nonzero integer n is a zero of multiplicity $1 + \nu_2(|n|)$, and

$$\Phi(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{1+\nu_2(n)}. \quad (21)$$

Gamma product (D). Euler’s reflection identity gives

$$\Phi(z) = \prod_{k=0}^{\infty} \frac{1}{\Gamma(1 + z/2^k) \Gamma(1 - z/2^k)}. \quad (22)$$

This is the natural entry point for the Barnes and multiple-gamma compressions studied in current repository drafts.

Spectral-zeta exponential (K). For $|z| < 1$,

$$\log \Phi(z) = - \sum_{m=1}^{\infty} \frac{\zeta(2m)}{m(1 - 2^{-2m})} z^{2m}. \quad (23)$$

The coefficient sequence is the spectral zeta function

$$Z_{\Phi}(s) = \sum_{n=1}^{\infty} \frac{1 + \nu_2(n)}{n^s} = \frac{\zeta(s)}{1 - 2^{-s}}. \quad (24)$$

Fredholm determinant (N). Section 9 upgrades (21) to

$$\Phi(z) = \det(I - z^2 \mathcal{K}), \quad (25)$$

for an explicit positive trace-class diagonal operator $\mathcal{K}$. This makes (24) literally the trace zeta function of $\mathcal{K}$.

3.3 Moment, cumulant, and Bell-polynomial representations

The moment generating function has the convergent Taylor series

$$M(t) = \sum_{n=0}^{\infty} \mu_{2n} \frac{t^{2n}}{(2n)!}. \quad (26)$$

Its logarithm is

$$\log M(t) = \sum_{m=1}^{\infty} \frac{2^{2m-1} B_{2m}}{m(2m)!(4^m - 1)} t^{2m}, \quad (27)$$

so all moments are complete Bell polynomials in explicit Bernoulli/Mersenne cumulants. These formulas, together with reciprocal Bell expansions for $1/\Phi$, are developed extensively in the repository.

3.4 Transform representations of F and F^{-1}

The CDF inversion formula gives

$$F(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^{\infty} \Phi(\xi) \frac{\sin(2\pi(2x-1)\xi)}{\xi} d\xi, \quad 0 < x < 1. \quad (28)$$

Section 7 adds closed Laplace and Fourier transforms of F as a function, as well as transforms of its derivative measure and its inverse.

3.5 A compact atlas table

Table 1: Representation atlas. Tags refer to the audited-corpus novelty boundary.

Object	Representation family	Tag	Location
ρ	fold of F , compact-support DDE, infinite convolution, random dyadic sum	K	§2–3
ρ	finite Thue–Morse truncated-power splines	K	(18)
ρ	Fourier inversion, cosine integral, Poisson and Legendre series	K	(19)
ρ	boundary jump of a finite logarithmic resolvent sum	N	Thm. 6.1
Φ	sinc product and weighted cosine product	K	(7)
Φ	canonical integer-zero product	K	(21)
Φ	gamma/Barnes products	K/D	(22)
Φ	zeta-exponential series	K	(23)
Φ	trace-class Fredholm determinant	N	Thm. 9.2
Φ'/Φ	cotangent dyadic sum	N	(60)
Φ'/Φ	weighted tangent dyadic sum	N	(61)
Φ'/Φ	valuation-weighted partial fractions	N	(62)
$(\log \Phi)^{(r)}$	digamma and Hurwitz-zeta hierarchy	N	Thm. 8.2

Object	Representation family	Tag	Location
M	hyperbolic-sinc product, moment series, Bell cumulants	K	(11)–(27)
M	positive Fredholm determinant	N	(71)
R	moment Laurent series	N	(31)
R	Laplace transform of the hyperbolic product	N	(32)
R	one-sided Fourier–Laplace integral	N	(33)
R	logarithmic fixed-point expectation	N	(36)
R	dyadic functional-differential equation	N	(37)
$R^{(n)}$	finite Thue–Morse affine orbit of R	N	(38)
R	finite 2^N -logarithm limit	N	(41)
R	Jacobi continued fraction	N	(82)
R	Gaussian-quadrature partial fractions	N	(91)
$\sum \mu_{2n} t^n$	Stieltjes continued fraction and product integral	N	(84)
F	CDF sine-product integral	D	(28)
F	exact Laplace transform on $[0, 1]$	N	(45)
F	exact Fourier transform on $[0, 1]$	N	(46)
$\rho _{[0,1]}$	exact one-sided Laplace/Fourier/sine transforms	N	Thm. 7.2
F^{-1}	pushforward expectation, Laplace, Fourier, and Cauchy transforms	N	Thm. 7.3

4 The Cauchy–Stieltjes transform

Definition 4.1. For $z \in \mathbb{C} \setminus [-1, 1]$, define

$$R(z) = \int_{-1}^1 \frac{\rho(x)}{z - x} dx = \mathbb{E} \frac{1}{z - X}. \quad (29)$$

The transform R is odd because ρ is even. It is an anti-Herglotz function: when $\operatorname{Im} z > 0$,

$$\operatorname{Im} R(z) = -\operatorname{Im} z \int_{-1}^1 \frac{\rho(x)}{|z - x|^2} dx < 0. \quad (30)$$

In particular, R has no zeros off the cut. For real $z > 1$, $R(z) > 0$; for $z < -1$, $R(z) < 0$.

Theorem 4.2 (Four complementary representations of R). *The following formulas hold.*

(i) For $|z| > 1$,

$$R(z) = \sum_{n=0}^{\infty} \frac{\mu_{2n}}{z^{2n+1}}. \quad (31)$$

(ii) For $\operatorname{Re} z > 1$,

$$R(z) = \int_0^{\infty} e^{-zt} \prod_{k=1}^{\infty} \frac{\sinh(t/2^k)}{t/2^k} dt. \quad (32)$$

(iii) For $\operatorname{Im} z > 0$,

$$R(z) = -2\pi i \int_0^{\infty} e^{2\pi i z \xi} \Phi(\xi) d\xi. \quad (33)$$

For $\operatorname{Im} z < 0$, take the conjugate counterpart.

(iv) For $-1 < x < 1$, the non-tangential boundary values satisfy

$$\rho(x) = -\frac{1}{\pi} \lim_{y \downarrow 0} \operatorname{Im} R(x + iy), \quad (34)$$

$$\operatorname{PV} \int_{-1}^1 \frac{\rho(t)}{x - t} dt = \lim_{y \downarrow 0} \operatorname{Re} R(x + iy) = 2\pi \int_0^{\infty} \Phi(\xi) \sin(2\pi x \xi) d\xi. \quad (35)$$

Proof. For $|z| > 1$, expand $(z - X)^{-1} = z^{-1} \sum_{m \geq 0} (X/z)^m$, use $|X| \leq 1$, and then symmetry. This gives (31).

For $\operatorname{Re} z > 1$,

$$\frac{1}{z - X} = \int_0^\infty e^{-(z-X)t} dt.$$

Fubini's theorem is justified because $X \leq 1$. Taking expectation and using (11) yields (32).

For $\operatorname{Im} z > 0$,

$$\frac{1}{z - x} = -2\pi i \int_0^\infty e^{2\pi i(z-x)\xi} d\xi.$$

The exponential damping from $\operatorname{Im} z > 0$, together with the rapid decay of Φ , permits Fubini and gives (33). Finally, Plemelj's boundary formula gives (34); taking real parts of (33) gives (35). $\square$

Remark 4.3. The four formulas emphasize different structures. The Laurent series exposes exact rational moments; the Laplace integral exposes the hyperbolic product; the Fourier–Laplace integral exposes spectral decay; and the boundary values recover both ρ and its Hilbert transform. Thus R is a single analytic object joining the moment, product, and inversion theories.

5 A dyadic functional equation for the resolvent

5.1 Logarithmic fixed-point identity

Theorem 5.1 (Logarithmic fixed point). *For real $z > 1$,*

$$R(z) = \int_{-1}^1 \rho(y) \log\left(\frac{2z + 1 - y}{2z - 1 - y}\right) dy. \quad (36)$$

The identity extends by analytic continuation to any simply connected subdomain of $\mathbb{C} \setminus [-1, 1]$, with the logarithm chosen by continuation from large positive z .

Proof. Use the fixed point (10) and condition on X' :

$$\begin{aligned} R(z) &= \mathbb{E} \left[\int_{-1/2}^{1/2} \frac{du}{z - u - X'/2} \right] \\ &= \mathbb{E} \log\left(\frac{z + 1/2 - X'/2}{z - 1/2 - X'/2}\right), \end{aligned}$$

which is (36) after multiplying numerator and denominator by 2. $\square$

5.2 Functional-differential equation

Theorem 5.2 (Resolvent equation). *For every $z \in \mathbb{C} \setminus [-1, 1]$,*

$$\boxed{R'(z) = 2(R(2z + 1) - R(2z - 1))}. \quad (37)$$

Proof. Differentiate (36) first for $z > 1$:

$$\begin{aligned} R'(z) &= \mathbb{E} \left(\frac{2}{2z + 1 - X'} - \frac{2}{2z - 1 - X'} \right) \\ &= 2(R(2z + 1) - R(2z - 1)). \end{aligned}$$

Both sides are holomorphic on $\mathbb{C} \setminus [-1, 1]$, so the identity theorem gives the stated continuation. $\square$

This is the transform-side analogue of the Rvachev equation (4). It is not merely a Fourier rephrasing: it acts on the spectral parameter of a Cauchy transform and therefore interacts directly with moments, Padé approximation, and orthogonal polynomials.

5.3 All-order Thue–Morse derivative tree

Theorem 5.3 (Thue–Morse derivative orbit). *For every integer $n \geq 0$,*

$$R^{(n)}(z) = 2^{n(n+1)/2} \sum_{j=0}^{2^n-1} \varepsilon_j R(2^n z + 2^n - 1 - 2j). \quad (38)$$

Consequently,

$$\int_{-1}^1 \frac{\rho(x)}{(z-x)^{n+1}} dx = \frac{(-1)^n 2^{n(n+1)/2}}{n!} \sum_{j=0}^{2^n-1} \varepsilon_j R(2^n z + 2^n - 1 - 2j). \quad (39)$$

Proof. The case $n = 0$ is trivial and $n = 1$ is (37). Assume the formula at order n . Differentiating contributes a factor 2^n , and then applying (37) contributes another factor 2. The exponent therefore changes by $n + 1$, from $n(n + 1)/2$ to $(n + 1)(n + 2)/2$.

For the shifts, write $a_{n,j} = 2^n - 1 - 2j$. The two children are

$$2a_{n,j} + 1 = 2^{n+1} - 1 - 2(2j), \quad 2a_{n,j} - 1 = 2^{n+1} - 1 - 2(2j + 1).$$

The signs obey $\varepsilon_{2j} = \varepsilon_j$ and $\varepsilon_{2j+1} = -\varepsilon_j$, exactly matching the subtraction in (37). This proves (38). Formula (39) follows from

$$R^{(n)}(z) = (-1)^n n! \int_{-1}^1 \frac{\rho(x)}{(z-x)^{n+1}} dx.$$

$\square$

Taking boundary imaginary parts in (38) recovers the familiar iterated derivative formula for ρ . The new point is that the same finite Thue–Morse orbit controls every higher Cauchy kernel, not only the density derivatives.

5.4 Uniqueness in the Stieltjes class

Proposition 5.4. *Among Cauchy transforms of probability measures supported on $[-1, 1]$, R is the unique solution of (37) satisfying*

$$R(z) = \frac{1}{z} + \mathcal{O}(z^{-3}) \quad (z \rightarrow \infty). \quad (40)$$

Proof. Let $G(z) = \sum_{r \geq 0} m_r z^{-r-1}$ be the Laurent expansion at infinity of any such Cauchy transform. The normalization gives $m_0 = 1$ and $m_1 = 0$. Substitute this expansion into (37). In the coefficient of z^{-r-2} , the contribution containing m_r is

$$-(r+1)m_r + \frac{r+1}{2^r} m_r = -(r+1)(1-2^{-r})m_r,$$

while every other contribution contains only $m_0, \dots, m_{r-2}$. Since the displayed factor is nonzero for $r \geq 1$, the differential equation determines all moments recursively. The transform R is one solution, so G and R have identical moments. Probability measures on the compact interval $[-1, 1]$ are determinate by their moments (equivalently, polynomials are dense in $C[-1, 1]$); hence the two measures and their Cauchy transforms coincide. $\square$

6 Finite logarithmic resolvents and the spline boundary jump

The finite random sum X_N has two exact representations: its density is a Thue–Morse spline, while its Cauchy transform is a Thue–Morse sum of logarithms. The second formula appears to be absent from the audited corpus.

Theorem 6.1 (Finite logarithmic formula). *Let*

$$R_N(z) = \mathbb{E} \frac{1}{z - X_N}, \quad X_N = \sum_{k=1}^N 2^{-k} U_k.$$

For $z \notin [-1 + 2^{-N}, 1 - 2^{-N}]$, with logarithms continued consistently from $z = +\infty$,

$$R_N(z) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \varepsilon_j(z + a_{N,j})^{N-1} \operatorname{Log}(z + a_{N,j}). \quad (41)$$

Moreover, $R_N \rightarrow R$ locally uniformly on $\mathbb{C} \setminus [-1, 1]$, so

$$R(z) = \lim_{N \rightarrow \infty} \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{j=0}^{2^N-1} \varepsilon_j(z + a_{N,j})^{N-1} \operatorname{Log}(z + a_{N,j}). \quad (42)$$

Finally, the upper boundary jump of (41) is exactly the spline (18):

$$\rho_N(x) = -\frac{1}{\pi} \operatorname{Im} R_N(x + i0). \quad (43)$$

Proof. For $a > 0$, convolution with the uniform density on $[-a, a]$ sends a Cauchy transform $G'(z)$ to

$$\frac{1}{2a} \int_{-a}^a G'(z - u) \, du = \frac{G(z + a) - G(z - a)}{2a}. \quad (44)$$

Start from the point mass at zero, whose Cauchy transform is $1/z$. An N -fold antiderivative is

$$H_N(z) = \frac{z^{N-1} \operatorname{Log} z}{(N-1)!}; \quad H_N^{(N)}(z) = \frac{1}{z}.$$

Applying (44) successively with $a_k = 2^{-k}$ gives

$$R_N(z) = \frac{1}{\prod_{k=1}^N 2a_k} \sum_{\sigma \in \{\pm 1\}^N} \left(\prod_{k=1}^N \sigma_k \right) H_N \left(z + \sum_{k=1}^N \sigma_k a_k \right).$$

Since

$$\frac{1}{\prod_{k=1}^N 2a_k} = 2^{N(N-1)/2},$$

and binary subsets identify the shifts and signs with $a_{N,j}$ and ε_j , this is (41).

The residual tail $X - X_N$ has support in $[-2^{-N}, 2^{-N}]$, so $X_N \rightarrow X$ almost surely and uniformly in support. The kernels $(z - x)^{-1}$ are uniformly continuous on compact subsets away from $[-1, 1]$; hence $R_N \rightarrow R$ locally uniformly.

For the boundary value, $\text{Im Log}(x + a + i0) = \pi$ when $x + a < 0$ and zero when $x + a > 0$. The alternating finite difference annihilates every polynomial of degree less than N , so the sum of the full powers $(x + a_{N,j})^{N-1}$ is zero. Replacing the negative-part sum by minus the positive-part sum yields exactly (18). This proves (43). $\square$

Remark 6.2 (Two complementary approximation geometries). The finite-convolution approximant R_N is a sum of logarithms and preserves the dyadic construction. The Jacobi approximants in Section 10 are rational functions and preserve moments. Both converge locally uniformly off the support, but their poles, branch cuts, and error mechanisms are completely different. This duality suggests hybrid algorithms: use finite convolution to resolve endpoint geometry and a short Jacobi fraction to compress the remaining moment error.

7 Exact transforms of F , one-sided up, and F^{-1}

7.1 The Laplace and Fourier transforms of F

Theorem 7.1 (Transform of the Fabius function). *For complex s , with the value at $s = 0$ understood by continuity,*

$$\boxed{\int_0^1 e^{-sx} F(x) \, dx = \frac{e^{-s/2} M(s/2) - e^{-s}}{s}.} \quad (45)$$

Equivalently, for real or complex ξ ,

$$\boxed{\int_0^1 F(x) e^{-2\pi i \xi x} \, dx = \frac{e^{-\pi i \xi} \Phi(\xi/2) - e^{-2\pi i \xi}}{2\pi i \xi}.} \quad (46)$$

At $\xi = 0$, the value is $1/2$.

Proof. The distributional Laplace transform of Y is

$$\int_0^1 e^{-sx} \, dF(x) = \mathbb{E} e^{-sY} = e^{-s/2} M(s/2). \quad (47)$$

Integration by parts gives

$$\int_0^1 e^{-sx} \, dF(x) = e^{-s} + s \int_0^1 e^{-sx} F(x) \, dx,$$

which is (45). Substitute $s = 2\pi i \xi$ and use $M(\pi i \xi) = \Phi(\xi/2)$. $\square$

Separating real and imaginary parts gives the explicit sine and cosine transforms

$$\int_0^1 F(x) \cos(2\pi \xi x) \, dx = \frac{\sin(2\pi \xi) - \Phi(\xi/2) \sin(\pi \xi)}{2\pi \xi}, \quad (48)$$

$$\int_0^1 F(x) \sin(2\pi \xi x) \, dx = \frac{\Phi(\xi/2) \cos(\pi \xi) - \cos(2\pi \xi)}{2\pi \xi}. \quad (49)$$

7.2 One-sided transforms of the up-function

On $[0, 1]$, $\rho(x) = F(1 - x) = 1 - F(x)$. Therefore the previous theorem immediately yields the following closed forms.

Theorem 7.2 (One-sided transform package). *For complex s , with the removable value at zero,*

$$\boxed{\int_0^1 e^{-sx} \rho(x) \, dx = \frac{1 - e^{-s/2} M(s/2)}{s}.} \quad (50)$$

Consequently,

$$\boxed{\int_0^1 e^{-2\pi i \xi x} \rho(x) \, dx = \frac{1 - e^{-\pi i \xi} \Phi(\xi/2)}{2\pi i \xi}.} \quad (51)$$

The half-line cosine and sine transforms are

$$\int_0^1 \rho(x) \cos(2\pi \xi x) \, dx = \frac{1}{2} \Phi(\xi), \quad (52)$$

$$\int_0^1 \rho(x) \sin(2\pi \xi x) \, dx = \frac{1 - \Phi(\xi/2) \cos(\pi \xi)}{2\pi \xi}. \quad (53)$$

The sum of (50) at s and $-s$ recovers $M(s)$ precisely because

$$M(s) = \frac{2 \sinh(s/2)}{s} M(s/2), \quad (54)$$

which is the hyperbolic continuation of (8).

7.3 The inverse Fabius function as a transport map

Let $I = F^{-1}$ on $[0, 1]$. Since I pushes Lebesgue measure forward to the law of Y , every integral of a composition with I reduces to the up-law.

Theorem 7.3 (Inverse pushforward master identity). *For every Borel function g for which one side is integrable,*

$$\boxed{\int_0^1 g(I(u)) \, du = \int_0^1 g(x) \, dF(x) = \int_{-1}^1 g\left(\frac{x+1}{2}\right) \rho(x) \, dx.} \quad (55)$$

In particular,

$$\int_0^1 e^{-sI(u)} \, du = e^{-s/2} M(s/2), \quad (56)$$

$$\int_0^1 e^{-2\pi i \xi I(u)} \, du = e^{-\pi i \xi} \Phi(\xi/2), \quad (57)$$

$$\int_0^1 \frac{du}{z - I(u)} = 2R(2z - 1), \quad z \in \mathbb{C} \setminus [0, 1], \quad (58)$$

$$\int_0^1 I(u)^n \, du = 2^{-n} \sum_{r=0}^{\lfloor n/2 \rfloor} \binom{n}{2r} \mu_{2r}. \quad (59)$$

Proof. If U is uniform on $[0, 1]$, then $I(U)$ has CDF F , hence the same law as Y . This proves (55). The remaining formulas follow by taking $g(x) = e^{-sx}$, $e^{-2\pi i \xi x}$, $(z - x)^{-1}$, and x^n , respectively, and then using (13) and symmetry of X . $\square$

Remark 7.4. Formula (58) transports the entire Jacobi-fraction theory of R to the inverse Fabius function. Thus a function whose pointwise inverse asymptotics involve Lambert W nevertheless has a globally simple Cauchy transform when sampled with uniform measure in its range variable.

8 Logarithmic derivatives of the Fourier image

8.1 Three meromorphic expansions

Theorem 8.1 (Cotangent, tangent, and zero-set formulas). *Away from the integer zero set of Φ ,*

$$\frac{\Phi'(z)}{\Phi(z)} = \sum_{k=0}^{\infty} \left[\frac{\pi}{2^k} \cot\left(\frac{\pi z}{2^k}\right) - \frac{1}{z} \right], \quad (60)$$

$$= -\pi \sum_{m=1}^{\infty} \frac{m}{2^m} \tan\left(\frac{\pi z}{2^m}\right), \quad (61)$$

$$= -2z \sum_{n=1}^{\infty} \frac{1 + \nu_2(n)}{n^2 - z^2}. \quad (62)$$

All three series converge locally normally on compact subsets avoiding the poles.

Proof. Differentiate the logarithm of each of the three products (7) and (21). For large k , the bracket in (60) and the term in (61) are $\mathcal{O}(4^{-k})$ uniformly on compact sets. The terms in (62) are bounded by a constant times $(1 + \nu_2(n))/n^2$, which is summable by (24) at $s = 2$. $\square$

The three forms have different numerical regimes. The tangent sum is usually the fastest away from its poles; the cotangent sum tracks the original sinc factors; and the zero-set sum exposes multiplicities and supports rigorous rational tail bounds.

8.2 Digamma and Hurwitz-zeta hierarchy

The valuation weight has the divisor decomposition

$$1 + \nu_2(n) = \sum_{k=0}^{\infty} \mathbf{1}_{2^k | n}. \quad (63)$$

This turns the zero sum into a dyadic superposition of classical special functions.

Theorem 8.2 (Digamma and Hurwitz-zeta representations). *Let $\psi = \Gamma'/\Gamma$. Then*

$$\frac{\Phi'(z)}{\Phi(z)} = \sum_{k=0}^{\infty} 2^{-k} \left[\psi\left(1 - \frac{z}{2^k}\right) - \psi\left(1 + \frac{z}{2^k}\right) \right]. \quad (64)$$

For every integer $r \geq 2$,

$$\frac{d^r}{dz^r} \log \Phi(z) = (r-1)! \sum_{k=0}^{\infty} 2^{-kr} \left[-\zeta\left(r, 1 - \frac{z}{2^k}\right) + (-1)^{r-1} \zeta\left(r, 1 + \frac{z}{2^k}\right) \right]. \quad (65)$$

Proof. From (21), for $r \geq 1$, paired symmetrically when $r = 1$,

$$(\log \Phi)^{(r)}(z) = (r-1)! \sum_{n \geq 1} (1 + \nu_2(n)) \left[-\frac{1}{(n-z)^r} + \frac{(-1)^{r-1}}{(n+z)^r} \right]. \quad (66)$$

Insert (63), write $n = 2^k m$, and sum over $m \geq 1$. For $r \geq 2$, the inner sums are Hurwitz zeta functions. For $r = 1$, the convergent paired sum is the standard digamma difference. $\square$

Corollary 8.3 (Local jet data at nonzero zeros). *Let $N \neq 0$ be an integer and $d_N = 1 + \nu_2(|N|)$. Then*

$$\frac{\Phi'(z)}{\Phi(z)} = \frac{d_N}{z-N} + H_N(z), \quad (67)$$

where H_N is holomorphic near N , and every derivative of H_N can be written as an absolutely convergent valuation-weighted Hurwitz-zeta sum. Thus the complete local Taylor jet of $\Phi(z)/(z-N)^{d_N}$ is algorithmically explicit.

9 A Fredholm determinant for the Fourier product

The canonical product can be realized as an ordinary Fredholm determinant, not merely as a formal spectral product.

Definition 9.1. Let

$$\mathcal{I} = \{(n, r) : n \geq 1, 1 \leq r \leq 1 + \nu_2(n)\}, \quad (68)$$

and define the positive diagonal operator $\mathcal{K}$ on $\ell^2(\mathcal{I})$ by

$$\mathcal{K}e_{n,r} = \frac{1}{n^2}e_{n,r}. \quad (69)$$

Theorem 9.2 (Fredholm determinant representation). *The operator $\mathcal{K}$ is trace class and*

$$\boxed{\Phi(z) = \det(I - z^2 \mathcal{K})}, \quad (70)$$

$$\boxed{M(t) = \det\left(I + \frac{t^2}{4\pi^2} \mathcal{K}\right)}. \quad (71)$$

Furthermore,

$$\mathrm{tr}(\mathcal{K}^m) = \frac{\zeta(2m)}{1 - 2^{-2m}}, \quad (72)$$

$$\frac{\Phi'(z)}{\Phi(z)} = -2z \mathrm{tr}(\mathcal{K}(I - z^2 \mathcal{K})^{-1}). \quad (73)$$

Proof. By (24),

$$\mathrm{tr} \mathcal{K} = \sum_{n \geq 1} \frac{1 + \nu_2(n)}{n^2} = \frac{\zeta(2)}{1 - 2^{-2}} < \infty,$$

so $\mathcal{K}$ is trace class. Its Fredholm determinant is the product over its eigenvalues, with multiplicity:

$$\det(I - z^2 \mathcal{K}) = \prod_{n \geq 1} \left(1 - \frac{z^2}{n^2}\right)^{1 + \nu_2(n)} = \Phi(z).$$

Substituting $z = it/(2\pi)$ gives (71). Formula (72) is again (24), and differentiating the Fredholm logarithm gives (73). $\square$

Remark 9.3. This representation converts several scalar identities into operator identities. For example, the zeta expansion (23) is the standard trace expansion

$$\log \det(I - z^2 \mathcal{K}) = - \sum_{m \geq 1} \frac{z^{2m}}{m} \operatorname{tr}(\mathcal{K}^m).$$

It also suggests perturbation theory for generalized exponent sequences: changing the multiplicity sequence changes only the diagonal spectral multiplicities.

10 Jacobi continued fractions and exact rational coefficients

10.1 Exact moment recurrence

Proposition 10.1 (Moment recurrence). *The even moments satisfy $\mu_0 = 1$ and, for $n \geq 1$,*

$$(4^n - 1)\mu_{2n} = \sum_{j=1}^n \binom{2n}{2j} \frac{\mu_{2n-2j}}{2j+1}. \quad (74)$$

In particular, every μ_{2n} is rational.

Proof. Raise (10) to the $2n$ -th power and take expectations:

$$4^n \mu_{2n} = \sum_{j=0}^n \binom{2n}{2j} \mathbb{E} U^{2j} \mu_{2n-2j}.$$

Since $\mathbb{E} U^{2j} = 1/(2j+1)$, the $j=0$ term is μ_{2n} . Move it to the left. $\square$

10.2 Monic orthogonal polynomials

Let P_n be the monic orthogonal polynomials for the probability measure $\rho(x) \, dx$, and let

$$h_n = \int_{-1}^1 P_n(x)^2 \rho(x) \, dx. \quad (75)$$

Evenness forces the diagonal Jacobi parameters to vanish, so

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_{n+1}(x) = xP_n(x) - \beta_n P_{n-1}(x), \quad \beta_n = \frac{h_n}{h_{n-1}} > 0. \quad (76)$$

The first polynomials are

$$P_0(x) = 1, \quad (77)$$

$$P_1(x) = x, \quad (78)$$

$$P_2(x) = x^2 - \frac{1}{9}, \quad (79)$$

$$P_3(x) = x^3 - \frac{19}{75}x, \quad (80)$$

$$P_4(x) = x^4 - \frac{62}{147}x^2 + \frac{619}{33075}. \quad (81)$$

10.3 The Jacobi and Stieltjes fractions

Theorem 10.2 (Jacobi continued fraction). *The Cauchy transform has the locally uniform continued fraction*

$$R(z) = \frac{1}{z - \frac{\beta_1}{z - \frac{\beta_2}{z - \frac{\beta_3}{z - \ddots}}}}. \quad (82)$$

The coefficients are positive rational numbers.

Proof. Positivity follows from $\beta_n = h_n/h_{n-1}$, and rationality follows because Gram–Schmidt applied to the rational moment functional (74) uses only rational operations. Markov’s theorem for compactly supported positive measures identifies the continued fraction generated by (76) with the Cauchy transform and gives local uniform convergence off the support [7, 6]. $\square$

Define the ordinary even-moment generating function

$$\mathcal{O}(t) = \sum_{n=0}^{\infty} \mu_{2n} t^n = \int_{-1}^1 \frac{\rho(x)}{1 - tx^2} dx, \quad |t| < 1. \quad (83)$$

Substituting $t = z^{-2}$ into (82) gives the contracted fraction.

Corollary 10.3 (Stieltjes fraction and product integral). *For $|t| < 1$,*

$$\mathcal{O}(t) = \frac{1}{1 - \frac{\beta_1 t}{1 - \frac{\beta_2 t}{1 - \frac{\beta_3 t}{1 - \ddots}}}}. \quad (84)$$

$$\mathcal{O}(t) = \int_0^{\infty} e^{-u} \prod_{k=1}^{\infty} \frac{\sinh(\sqrt{t} u/2^k)}{\sqrt{t} u/2^k} du. \quad (85)$$

Proof. The continued fraction follows from $\mathcal{O}(z^{-2}) = zR(z)$. For the integral, use

$$\frac{1}{1 - tx^2} = \int_0^{\infty} e^{-u} \cosh(\sqrt{t} xu) du$$

and then average over X , obtaining $M(\sqrt{t} u)$. $\square$

For $s \geq 0$, $\mathcal{O}(-s)$ is completely monotone:

$$(-1)^m \frac{d^m}{ds^m} \mathcal{O}(-s) = m! \int_{-1}^1 \frac{x^{2m} \rho(x)}{(1 + sx^2)^{m+1}} dx > 0. \quad (86)$$

Thus the even moments form a Stieltjes moment sequence, and the fraction (84) belongs to the classical positive Stieltjes class.

10.4 Hankel determinants

Let

$$\Delta_n = \det[\mu_{i+j}]_{i,j=0}^{n-1}, \quad \Delta_0 = 1. \quad (87)$$

Then

$$h_n = \frac{\Delta_{n+1}}{\Delta_n}, \quad \boxed{\beta_n = \frac{\Delta_{n+1}\Delta_{n-1}}{\Delta_n^2}}. \quad (88)$$

Because odd moments vanish, each Δ_n factors after a parity permutation into an even-even and odd-odd moment determinant. This parity factorization is useful for exact arithmetic and should be exploited in future Lean formalization.

10.5 Exact coefficients and numerical evidence

The accompanying Python program computes the moments by (74) and then uses the exact symmetric Stieltjes procedure (76). Selected results are:

n	μ_{2n}	decimal
0	1	1
1	$\frac{1}{9}$	0.111111111111
2	$\frac{19}{675}$	0.0281481481481
3	$\frac{583}{59535}$	0.00979255899891
4	$\frac{132809}{32531625}$	0.00408245822334
5	$\frac{46840699}{24405225075}$	0.00191928977733
6	$\frac{4068990560161}{4133856862760625}$	0.000984308527181
7	$\frac{1204567303451311}{2232691548877164375}$	0.000539513532022
8	$\frac{4146897304424408411}{13301767332333178846875}$	0.00031175536309
9	$\frac{18814360006695807527868793}{100028040755473167511640090625}$	0.000188090857969
10	$\frac{21431473463327429953796293981397}{182171989134769427819794434994453125}$	0.000117644175513

n	β_n (exact when compact)	decimal
1	$\frac{1}{9}$	0.111111111111
2	$\frac{32}{225}$	0.142222222222
3	$\frac{619}{3675}$	0.168435374150
4	$\frac{835232}{4640643}$	0.179981955087
5	$\frac{1789043898625}{9272274723657}$	0.192945523288
6	see data file	0.199759514028
7	see data file	0.205964295192
8	see data file	0.211147954951
9	see data file	0.215261988744
10	see data file	0.218406233337
11	see data file	0.221259971300
12	see data file	0.223692820062

The full numerators and denominators through $n = 40$ are in `up_jacobi_coefficients.csv`. Their size grows rapidly: by moderate order the reduced integers contain thousands of decimal digits. This arithmetic complexity is not a floating-point artifact.

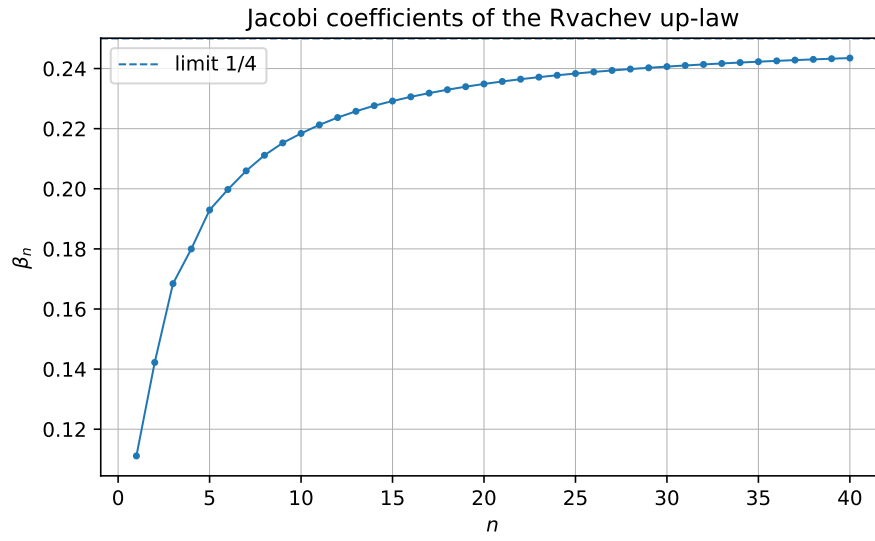


Figure 1: Exact Jacobi coefficients converted to decimals. The dashed line is the proved limit $1/4$.

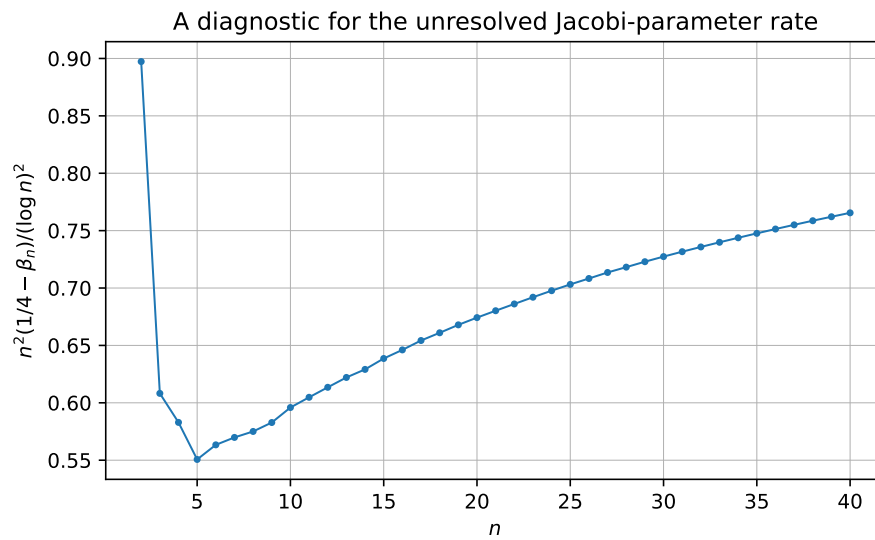


Figure 2: The diagnostic quantity $n^2(1/4 - \beta_n)/(\log n)^2$. The range $n \leq 40$ is evidence, not an asymptotic proof.

10.6 The limiting coefficient

Theorem 10.4 (Nevai-class limit). *The Jacobi coefficients satisfy*

$$\boxed{\lim_{n \rightarrow \infty} \beta_n = \frac{1}{4}}. \quad (89)$$

Proof. The measure $\rho(x) dx$ has essential support $[-1, 1]$, and its density is strictly positive for every $x \in (-1, 1)$. The Denisov–Rakhmanov theorem therefore places its orthonormal Jacobi matrix in the Nevai class of $[-1, 1]$ [3, 6]: the off-diagonal orthonormal coefficients tend to $1/2$, and the diagonal coefficients tend to zero. In the monic recurrence (76), β_n is the square of the off-diagonal orthonormal coefficient, hence tends to $1/4$. $\square$

10.7 Padé and Gaussian-quadrature representations

Let Q_{N-1} be the polynomial of the second kind

$$Q_{N-1}(z) = \int_{-1}^1 \frac{P_N(z) - P_N(x)}{z - x} \rho(x) dx. \quad (90)$$

The N -th Jacobi convergent is Q_{N-1}/P_N . If $x_{1,N}, \dots, x_{N,N}$ are the zeros of P_N , then

$$\boxed{\frac{Q_{N-1}(z)}{P_N(z)} = \sum_{j=1}^N \frac{w_{j,N}}{z - x_{j,N}}, \quad w_{j,N} > 0, \quad \sum_j w_{j,N} = 1.} \quad (91)$$

This is the Cauchy transform of the N -node Gaussian quadrature measure for ρ , and it matches moments through degree $2N - 1$. The exact error identity is

$$R(z) - \frac{Q_{N-1}(z)}{P_N(z)} = \frac{1}{P_N(z)^2} \int_{-1}^1 \frac{P_N(x)^2}{z - x} \rho(x) dx. \quad (92)$$

Hence every Gaussian resolvent approximant is a strict lower bound for $R(z)$ when $z > 1$, and a strict upper bound when $z < -1$.

The first nontrivial approximants are completely explicit:

$$R_1^G(z) = \frac{1}{z}, \quad (93)$$

$$R_2^G(z) = \frac{z}{z^2 - 1/9} = \frac{1}{2} \left(\frac{1}{z - 1/3} + \frac{1}{z + 1/3} \right), \quad (94)$$

$$R_3^G(z) = \frac{32/57}{z} + \frac{25/114}{z - \sqrt{19/75}} + \frac{25/114}{z + \sqrt{19/75}}. \quad (95)$$

Thus the continued fraction gives a hierarchy of positive finite atomic models for the up-law, complementary to the continuous finite-convolution models of Section 5.

11 Further combined representations

11.1 The Cauchy transform of the Fabius measure

Let

$$S(z) = \int_0^1 \frac{dF(x)}{z - x} = \mathbb{E} \frac{1}{z - Y}. \quad (96)$$

The affine relation (13) gives

$$\boxed{S(z) = 2R(2z - 1).} \quad (97)$$

Consequently, every representation of R immediately yields one for S . For example, for $|2z - 1| > 1$,

$$S(z) = 2 \sum_{n=0}^{\infty} \frac{\mu_{2n}}{(2z - 1)^{2n+1}}, \quad (98)$$

and S has a Jacobi fraction obtained by the affine change of variable in (82).

11.2 A determinant form of the Fabius Laplace transform

Combining (45) with (71) gives

$$\int_0^1 e^{-sx} F(x) \, dx = \frac{e^{-s/2} \det \left(I + \frac{s^2}{16\pi^2} \mathcal{K} \right) - e^{-s}}{s}. \quad (99)$$

Similarly,

$$\int_0^1 e^{-sx} \rho(x) \, dx = \frac{1 - e^{-s/2} \det \left(I + \frac{s^2}{16\pi^2} \mathcal{K} \right)}{s}. \quad (100)$$

These are literal combinations of a compact-support transform with a trace-class spectral determinant.

11.3 A resolvent–product–continued-fraction triangle

For $|t| < 1$, the same analytic function $\mathcal{O}(t)$ has three structurally different representations:

$$\boxed{\begin{aligned} \mathcal{O}(t) &= \sum_{n \geq 0} \mu_{2n} t^n \\ &= \int_0^\infty e^{-u} \prod_{k \geq 1} \frac{\sinh(\sqrt{t} u / 2^k)}{\sqrt{t} u / 2^k} \, du \\ &= \frac{1}{1 - \frac{\beta_1 t}{1 - \frac{\beta_2 t}{1 - \ddots}}}. \end{aligned}} \quad (101)$$

The first line is arithmetic, the second is multiplicative and analytic, and the third is orthogonal-polynomial/geometric. Equality of the three creates a direct bridge among the Bernoulli–Bell moment formulas, the infinite hyperbolic product, and the exact rational Jacobi parameters.

12 Half-lattice Shannon reconstruction and a convergent Mittag–Leffler formula

The corpus already proves that the integer zeros of Φ and the signs of its half-integer samples are controlled by binary arithmetic. A different question is whether those samples reconstruct

the whole Fourier image. Compact support supplies an exact answer. The result is especially sparse because all nonzero integer samples vanish.

For this section write

$$c_k := \Phi\left(k + \frac{1}{2}\right), \quad a_k := 2k + 1, \quad A_k := |c_k| \quad (k \geq 0). \quad (102)$$

The half-integer sign theorem in the audited corpus gives

$$c_k = \varepsilon_k A_k, \quad \varepsilon_k = (-1)^{\text{wt}_2(k)}, \quad A_k > 0. \quad (103)$$

Because $\rho \in C_c^\infty([-1, 1])$, integration by parts shows that $c_k = \mathcal{O}_N(k^{-N})$ for every N . Every series below therefore converges absolutely, together with all termwise derivatives, on compact sets away from displayed poles.

Theorem 12.1 (Sparse half-lattice sampling). *For every $z \in \mathbb{C}$,*

$$\Phi(z) = \text{sinc}_\pi(2z) + \sum_{k=0}^{\infty} c_k \{ \text{sinc}_\pi(2z - a_k) + \text{sinc}_\pi(2z + a_k) \}. \quad (104)$$

The series converges locally uniformly on $\mathbb{C}$, hence represents Φ as an entire function and may be differentiated term by term to arbitrary order.

Proof. With the Fourier convention (5), the inverse Fourier transform of the entire function Φ is supported in $[-1, 1]$. Thus Φ lies in the Paley–Wiener space of bandwidth one. The Whittaker–Shannon theorem [11] at the critical spacing $1/2$ gives

$$\Phi(z) = \sum_{n \in \mathbb{Z}} \Phi(n/2) \text{sinc}_\pi(2z - n). \quad (105)$$

Here $\Phi(0) = 1$, while $\Phi(n) = 0$ for every nonzero integer n because the first factor $\text{sinc}_\pi(z)$ in (7) vanishes. The surviving indices are therefore $n = 0$ and the odd integers. Since Φ is even, $\Phi(-(2k + 1)/2) = c_k$, and pairing positive and negative odd indices yields (104). Rapid decrease of the samples upgrades the usual Paley–Wiener convergence to absolute local uniform convergence. $\square$

The paired cardinal kernel has a rational form:

$$\text{sinc}_\pi(2z - a) + \text{sinc}_\pi(2z + a) = -\frac{4z \sin(2\pi z)}{\pi(4z^2 - a^2)} \quad (a \in 2\mathbb{Z} + 1). \quad (106)$$

Substitution gives a genuine, rapidly convergent partial-fraction expansion.

Corollary 12.2 (Half-lattice Mittag–Leffler expansion). *As an identity of meromorphic functions,*

$$\frac{\pi\Phi(z)}{\sin(2\pi z)} = \frac{1}{2z} - 4z \sum_{k=0}^{\infty} \frac{c_k}{4z^2 - a_k^2}. \quad (107)$$

The only poles of the right-hand side are simple poles at the nonzero half-integers; at $z = k + \frac{1}{2}$,

$$\text{Res}_{z=k+1/2} \frac{\pi\Phi(z)}{\sin(2\pi z)} = -\frac{c_k}{2}. \quad (108)$$

Consequently the signed and unsigned Thue–Morse sequences admit the residue representations

$$\varepsilon_k = -\text{sgn} \left(\text{Res}_{z=k+1/2} \frac{\pi\Phi(z)}{\sin(2\pi z)} \right), \quad (109)$$

$$\tau_k = \frac{1}{2} \left[1 + \text{sgn} \left(\text{Res}_{z=k+1/2} \frac{\pi\Phi(z)}{\sin(2\pi z)} \right) \right]. \quad (110)$$

Remark 12.3 (Why this differs from the zero-set expansion). The audited corpus and the first-wave report contain valuation-weighted partial fractions for Φ'/Φ , whose poles are the integer zeros of Φ . Formula (107) is structurally different: it reconstructs Φ itself from its nonzero half-lattice samples, and its poles are introduced by division by $\sin(2\pi z)$. The residues carry the Thue–Morse signs rather than the integer-zero multiplicities.

Corollary 12.4 (Derivative cardinal series). *For every integer $r \geq 0$,*

$$\Phi^{(r)}(z) = 2^r \operatorname{sinc}_\pi^{(r)}(2z) + 2^r \sum_{k \geq 0} c_k \left\{ \operatorname{sinc}_\pi^{(r)}(2z - a_k) + \operatorname{sinc}_\pi^{(r)}(2z + a_k) \right\}. \quad (111)$$

In particular, every Taylor coefficient and every moment of the up-law can be recovered from the single signed sample sequence $(c_k)_{k \geq 0}$.

13 Odd-harmonic cosine series and spectral Prouhet cancellation

Sampling in frequency has a dual real-space formulation. Periodize ρ with period 2. Its Fourier coefficient at the integer frequency n is $\frac{1}{2}\Phi(n/2)$. The even frequencies disappear, leaving only odd harmonics.

Theorem 13.1 (Sparse Fourier series for ρ and F). *For $-1 \leq x \leq 1$,*

$$\rho(x) = \frac{1}{2} + \sum_{k=0}^{\infty} c_k \cos(a_k \pi x). \quad (112)$$

For $0 \leq x \leq 1$,

$$F(x) = \frac{1}{2} - \sum_{k=0}^{\infty} c_k \cos(a_k \pi x). \quad (113)$$

Both series converge absolutely in every C^r -norm. More explicitly, for $r \geq 0$,

$$\rho^{(2r)}(x) = (-1)^r \pi^{2r} \sum_{k \geq 0} c_k a_k^{2r} \cos(a_k \pi x), \quad (114)$$

$$\rho^{(2r+1)}(x) = (-1)^{r+1} \pi^{2r+1} \sum_{k \geq 0} c_k a_k^{2r+1} \sin(a_k \pi x). \quad (115)$$

The corresponding formulas for F differ only by the sign visible in (113).

Proof. The period-two Fourier series of ρ is

$$\rho(x) = \frac{1}{2} \sum_{n \in \mathbb{Z}} \Phi(n/2) e^{\pi i n x}.$$

As in the proof of Theorem 12.1, the only nonzero even-frequency coefficient is the constant coefficient. Pairing the odd positive and negative frequencies gives (112). On $[0, 1]$, the folding relation $\rho(x) = F(1-x) = 1 - F(x)$ gives (113). Rapid decay of c_k justifies all derivatives. $\square$

Formula (112) converts flatness at the center into infinitely many exact weighted Thue–Morse cancellations.

Corollary 13.2 (Spectral Prouhet identities). *With $A_k = |c_k|$,*

$$\boxed{\sum_{k=0}^{\infty} \varepsilon_k A_k = \frac{1}{2}, \quad \sum_{k=0}^{\infty} \varepsilon_k A_k a_k^{2r} = 0 \quad (r \geq 1).} \quad (116)$$

All sums are absolutely convergent. Equivalently, for $r \geq 1$, the unsigned sequence satisfies

$$\sum_{k \geq 0} \tau_k A_k a_k^{2r} = \frac{1}{2} \sum_{k \geq 0} A_k a_k^{2r}. \quad (117)$$

Proof. At $x = 0$, $\rho(0) = 1$, while $\rho^{(m)}(0) = 0$ for every $m \geq 1$: the up-function has a flat maximum at the origin. Set $x = 0$ in (112) and (114). Then use $c_k = \varepsilon_k A_k$. Finally substitute $\varepsilon_k = 1 - 2\tau_k$ to obtain (117). $\square$

These identities resemble the finite Prouhet cancellation $\sum_{n < 2^m} \varepsilon_n n^r = 0$ for $r < m$, but the mechanism and weights are different. Here one fixed infinite Thue–Morse word cancels *every* positive even moment after being weighted by the half-lattice Fourier amplitudes A_k . The weights themselves encode the smooth compactly supported fixed point.

Corollary 13.3 (Exact dyadic cosine-product sums). *For integers $m \geq 0$ and $0 \leq a \leq 2^m$,*

$$F\left(\frac{a}{2^m}\right) = \frac{1}{2} - \sum_{k=0}^{\infty} \varepsilon_k A_k \cos\left(\frac{a(2k+1)\pi}{2^m}\right). \quad (118)$$

The left side is rational by the exact dyadic arithmetic developed in the repository. Thus (118) is an infinite trigonometric-product representation of a rational number, with

$$A_k = \left| \prod_{j=0}^{\infty} \operatorname{sinc}_{\pi}\left(\frac{k+1/2}{2^j}\right) \right|. \quad (119)$$

A particularly direct new representation of both Thue–Morse forms follows from the Fourier coefficient itself:

$$\varepsilon_k = \operatorname{sgn} \int_{-1}^1 \rho(x) \cos(a_k \pi x) \, dx, \quad (120)$$

$$\tau_k = \frac{1}{2} \left(1 - \operatorname{sgn} \int_{-1}^1 \rho(x) \cos(a_k \pi x) \, dx \right). \quad (121)$$

The significance is not merely that a sign can be hidden inside an integral: the integrand uses the unique smooth dyadic probability fixed point, and the entire sign word is recovered from its odd Fourier spectrum.

14 Bose–Malmsten product-integrals for the Fourier image

The infinite sinc product can be converted into an integral over the classical Bose kernel $(e^t - 1)^{-1}$. Define the dyadic hyperbolic kernel

$$H(u) := \sum_{j=0}^{\infty} \left(\cosh(u/2^j) - 1 \right). \quad (122)$$

It is entire and obeys

$$H(u) = \cosh u - 1 + H(u/2), \quad (123)$$

$$H(u) = \sum_{r=1}^{\infty} \frac{u^{2r}}{(2r)!(1 - 4^{-r})}. \quad (124)$$

Theorem 14.1 (Bose–Malmsten representation). *In the zero-free strip $|\Re z| < 1$, with the branch normalized by $\log \Phi(0) = 0$,*

$$\log \Phi(z) = -2 \int_0^\infty \frac{H(z t)}{t(e^t - 1)} dt. \quad (125)$$

Equivalently,

$$\Phi(z) = \exp \left\{ -2 \int_0^\infty \frac{H(z t)}{t(e^t - 1)} dt \right\}. \quad (126)$$

Proof. For $|\Re w| < 1$, Euler’s product for the sine and Frullani’s integral give

$$\begin{aligned} \log \operatorname{sinc}_\pi(w) &= \sum_{n=1}^\infty \log \left(1 - \frac{w^2}{n^2} \right) \\ &= -2 \int_0^\infty \frac{\cosh(wt) - 1}{t(e^t - 1)} dt. \end{aligned}$$

Indeed, $\int_0^\infty e^{-nt} (\cosh(wt) - 1) dt/t = -\frac{1}{2} \log(1 - w^2/n^2)$. Sum this identity over $w = z/2^j$. Absolute convergence in compact substrips permits Fubini’s theorem, and the sum inside the integral is precisely $H(z t)$. $\square$

Corollary 14.2 (All logarithmic derivatives). *For $q \geq 1$ and $|\Re z| < 1$,*

$$\frac{d^q}{dz^q} \log \Phi(z) = -2 \int_0^\infty \frac{t^{q-1}}{e^t - 1} \sum_{j=0}^\infty 2^{-jq} \chi_q(z t/2^j) dt, \quad (127)$$

where $\chi_q = \sinh$ for odd q and $\chi_q = \cosh$ for even q . In particular,

$$\left. \frac{d^{2r}}{dz^{2r}} \log \Phi(z) \right|_{z=0} = -\frac{2(2r-1)! \zeta(2r)}{1 - 4^{-r}}. \quad (128)$$

Expanding H and using $\int_0^\infty t^{s-1}/(e^t - 1) dt = \Gamma(s)\zeta(s)$ yields the local zeta series

$$\log \Phi(z) = -\sum_{r=1}^\infty \frac{\zeta(2r)}{r(1 - 4^{-r})} z^{2r}, \quad |z| < 1. \quad (129)$$

This recovers the cumulant/Bernoulli expansion from a single positive-kernel integral and makes complete monotonicity on the imaginary axis transparent.

The strip restriction can be removed by retaining finitely many sinc factors.

Corollary 14.3 (Global finite-product–integral hybrid). *For any $z \in \mathbb{C}$, choose an integer $M \geq 0$ such that $|\Re(z/2^M)| < 1$. Then*

$$\Phi(z) = \left[\prod_{j=0}^{M-1} \operatorname{sinc}_\pi(z/2^j) \right] \exp \left\{ -2 \int_0^\infty \frac{H(z t/2^M)}{t(e^t - 1)} dt \right\}. \quad (130)$$

If a finite factor vanishes, both sides are interpreted by continuity and equal zero. For $z = k + \frac{1}{2}$, this gives a finite-product–integral representation of every spectral weight c_k , and hence of every sign in (109).

Combining (130) with Fourier inversion gives a nested integral representation of the up-function:

$$\rho(x) = \int_{\mathbb{R}} e^{2\pi ixt} \left[\prod_{j=0}^{M(t)-1} \operatorname{sinc}_{\pi}(t/2^j) \right] \exp \left\{ -2 \int_0^{\infty} \frac{H(tu/2^{M(t)})}{u(e^u - 1)} du \right\} dt, \quad (131)$$

where any measurable integer choice $M(t) > \log_2^+ |t|$ is admissible. Although this form is not intended for routine computation, it is an explicit integral–finite-product–exponential combination of the type requested in the present investigation.

15 Mellin and quantile representations

Let X have density ρ , let $Y = (1 + X)/2$, and let $I = F^{-1}$. Write

$$\mathcal{M}_Y(s) := \mathbb{E}Y^s = \int_0^1 y^s dF(y). \quad (132)$$

The endpoint flatness of F' implies that $y^{-A}F'(y)$ is integrable for every real A . Hence $\mathcal{M}_Y$ is an entire function of s , not merely a right-half-plane Mellin transform.

Theorem 15.1 (Mellin–quantile master package). *For all $s \in \mathbb{C}$,*

$$\boxed{\mathcal{M}_Y(s) = \int_0^1 I(u)^s du.} \quad (133)$$

For $\Re s > 0$,

$$\boxed{\int_0^1 x^{s-1} \rho(x) dx = \frac{\mathcal{M}_Y(s)}{s},} \quad (134)$$

$$\boxed{\int_0^1 x^{s-1} F(x) dx = \frac{1 - \mathcal{M}_Y(s)}{s}.} \quad (135)$$

The second right-hand side extends to an entire function at $s = 0$; the first has the expected simple pole of residue one there.

Proof. Equation (133) is the inverse-pushforward identity (55) with $g(y) = y^s$. Its convergence for arbitrary complex s follows from the flat density at zero. Since $\rho(x) = 1 - F(x)$ on $[0, 1]$, integration by parts gives

$$\int_0^1 x^{s-1} F(x) dx = \frac{1}{s} - \frac{1}{s} \int_0^1 x^s dF(x) = \frac{1 - \mathcal{M}_Y(s)}{s}.$$

Subtracting from $\int_0^1 x^{s-1} dx = 1/s$ gives (134). □

Let $\mu_{2j} = \mathbb{E}X^{2j}$ be the even up-moments from (12).

Theorem 15.2 (Generalized binomial moment series). *For $\Re s > 0$,*

$$\boxed{\mathcal{M}_Y(s) = 2^{-s} \sum_{j=0}^{\infty} \binom{s}{2j} \mu_{2j}.} \quad (136)$$

The series is absolutely convergent. For a nonnegative integer n , it terminates and reduces to

$$\mathcal{M}_Y(n) = 2^{-n} \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{2j} \mu_{2j}. \quad (137)$$

Moreover,

$$\mathcal{M}_Y(s) = 2^{-s} \sum_{j=0}^{\infty} \binom{s}{2j} \frac{(-1)^j \Phi^{(2j)}(0)}{(2\pi)^{2j}}. \quad (138)$$

Proof. For $\Re s > 0$, the generalized binomial coefficients are absolutely summable, and $(1 + X)^s = \sum_{n \geq 0} \binom{s}{n} X^n$ may be averaged termwise because $|X| \leq 1$. Symmetry kills the odd moments. The Fourier-jet form follows from $\Phi^{(2j)}(0) = (-1)^j (2\pi)^{2j} \mu_{2j}$. $\square$

Differentiation at $s = 0$ supplies logarithmic inverse moments that are difficult to see pointwise:

$$\mathcal{M}_Y^{(r)}(0) = \int_0^1 \log^r I(u) \, du = \int_0^1 \log^r y \, dF(y) \quad (r \geq 0). \quad (139)$$

Thus every logarithmic moment of the inverse Fabius function is a derivative of one entire Mellin function. Together with the Bell-polynomial formulas for μ_{2j} , (136) forms a Mellin–Bell–Fourier representation chain.

16 Formal Thue–Morse orthogonality and nonlinear Jacobi renormalization

The signed Thue–Morse generating function

$$E(z) := \sum_{n=0}^{\infty} \varepsilon_n z^n = \prod_{j=0}^{\infty} (1 - z^{2^j}), \quad |z| < 1, \quad (140)$$

can itself be regarded as a moment generating function, but for an indefinite formal functional rather than a positive measure. Define

$$\mathcal{L}[x^n] := \varepsilon_n. \quad (141)$$

The nonvanishing Hankel theorem of Allouche–Peyrière–Wen–Wen [9, 10] guarantees that $\mathcal{L}$ is quasi-definite: there is a unique monic formal orthogonal polynomial p_n of every degree. Put

$$h_n := \mathcal{L}[p_n^2], \quad (142)$$

$$p_{n+1}(x) = (x - a_n)p_n(x) - b_n p_{n-1}(x), \quad (143)$$

$$p_{-1} = 0, \quad p_0 = 1, \quad b_n = h_n/h_{n-1} \quad (n \geq 1). \quad (144)$$

Unlike the positive Jacobi matrix of Section 10, the numbers b_n here may have either sign.

The binary recursion gives two identities valid for every polynomial f :

$$\mathcal{L}[f(x^2)] = \mathcal{L}[f(x)], \quad \mathcal{L}[xf(x^2)] = -\mathcal{L}[f(x)]. \quad (145)$$

They force a complete renormalization law.

Theorem 16.1 (Quadratic decimation of the formal orthogonal polynomials). *For every $n \geq 0$,*

$$p_{2n}(x) = p_n(x^2), \quad (146)$$

$$h_{2n} = h_n, \quad (147)$$

$$a_n = (-1)^{n+1}. \quad (148)$$

Every odd polynomial has a factor $x + 1$: there is a monic polynomial q_n such that

$$p_{2n+1}(x) = (x + 1)q_n(x^2). \quad (149)$$

The off-diagonal coefficients are determined from $b_1 = -2$ by the nonlinear dyadic recursion

$$b_{2n-1}b_{2n} = b_n, \quad (150)$$

$$b_{2n} + b_{2n+1} = (-1)^{n+1} - 1 \quad (n \geq 1). \quad (151)$$

Equivalently,

$$b_{2n} = \frac{b_n}{b_{2n-1}}, \quad b_{2n+1} = (-1)^{n+1} - 1 - b_{2n}. \quad (152)$$

Proof. The polynomial $p_n(x^2)$ is monic of degree $2n$. For $j < n$,

$$\begin{aligned} \mathcal{L}[p_n(x^2)x^{2j}] &= \mathcal{L}[p_n(x)x^j] = 0, \\ \mathcal{L}[p_n(x^2)x^{2j+1}] &= -\mathcal{L}[p_n(x)x^j] = 0, \end{aligned}$$

by (145). These are all monomials of degree below $2n$, so uniqueness of the monic orthogonal polynomial proves (146). The norm identity follows immediately.

For even indices,

$$a_{2n} = \frac{\mathcal{L}[xp_{2n}^2]}{h_{2n}} = -\frac{\mathcal{L}[p_n^2]}{h_n} = -1.$$

Since $a_0 = -1$, $p_1 = x + 1$. The recurrence at an even index now shows inductively that p_{2n+1} is divisible by $x + 1$, proving (149). Write $p_{2n+1} = (x + 1)q_n(x^2)$ and set $f = q_n^2$. The moment rules imply

$$\begin{aligned} \mathcal{L}[(x + 1)^2 f(x^2)] &= \mathcal{L}[xf(x)] - \mathcal{L}[f(x)], \\ \mathcal{L}[x(x + 1)^2 f(x^2)] &= \mathcal{L}[xf(x)] - \mathcal{L}[f(x)]. \end{aligned}$$

Their quotient is one, so $a_{2n+1} = 1$. This proves (148).

Finally use the recurrences at indices $2n - 1, 2n, 2n + 1$. From

$$p_{2n} = (x - 1)p_{2n-1} - b_{2n-1}p_{2n-2}$$

we obtain

$$(x - 1)p_{2n-1} = p_n(x^2) + b_{2n-1}p_{n-1}(x^2).$$

Substitute this and $p_{2n+1} = (x + 1)p_n(x^2) - b_{2n}p_{2n-1}$ into the recurrence for p_{2n+2} . The result is

$$p_{2n+2} = [x^2 - 1 - b_{2n} - b_{2n+1}]p_n(x^2) - b_{2n-1}b_{2n}p_{n-1}(x^2).$$

Compare with $p_{n+1}(x^2) = [x^2 - a_n]p_n(x^2) - b_n p_{n-1}(x^2)$ to obtain (150)–(151). Directly, $b_1 = h_1/h_0 = \mathcal{L}[(x + 1)^2] = -2$. $\square$

The first coefficients are

$$\begin{array}{c|cccccccccccccccc} n & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 & 11 & 12 & 13 & 14 & 15 & 16 \\ \hline b_n & -2 & 1 & -1 & -1 & -1 & 1 & -1 & 1 & -3 & \frac{1}{3} & -\frac{1}{3} & -3 & 1 & -1 & 1 & 1. \end{array} \quad (153)$$

The nonlinear recursion computes the whole sequence using rational arithmetic without any Hankel determinant. It also proves that the formal Jacobi fraction of E has exactly alternating diagonal entries:

$$E(z) = \frac{1}{1 + z - \frac{b_1 z^2}{1 - z - \frac{b_2 z^2}{1 + z - \frac{b_3 z^2}{1 - z - \ddots}}}}. \quad (154)$$

The equality is an identity of formal power series and hence analytic for sufficiently small $|z|$.

Corollary 16.2 (Dyadic formal orthogonal polynomials). *For $N = 2^m$,*

$$\boxed{p_N(x) = x^N + 1, \quad h_N = -2.} \quad (155)$$

Proof. Iterate (146) from $p_1 = x + 1$; iterate (147) from $h_1 = -2$. $\square$

17 Explicit dyadic Padé approximants and antiperiodic quadrature

Let

$$P_m(z) := \sum_{n=0}^{2^m-1} \varepsilon_n z^n = \prod_{j=0}^{m-1} (1 - z^{2^j}), \quad N := 2^m. \quad (156)$$

The Mahler/block factorization is $E(z) = P_m(z)E(z^N)$. The elementary approximation $E(w) = 1 - w - w^2 + \mathcal{O}(w^3)$ has a surprisingly strong dyadic lift.

Theorem 17.1 (Closed dyadic Padé approximant). *For every $m \geq 0$,*

$$\boxed{E(z) - \frac{P_m(z)}{1 + z^N} = -2z^{2N} + \mathcal{O}(z^{2N+1}).} \quad (157)$$

Thus $P_m(z)/(1 + z^N)$ is the Padé approximant of type $[N - 1/N]$ to $E(z)$. The exact error identity is

$$E(z) - \frac{P_m(z)}{1 + z^N} = P_m(z) \left(E(z^N) - \frac{1}{1 + z^N} \right). \quad (158)$$

In the standard Padé–orthogonality correspondence [12, 13], the denominator $1 + z^N$ is the reciprocal polynomial of the formal orthogonal polynomial $p_N(x) = x^N + 1$ in (155).

Proof. Use $E(z) = P_m(z)E(z^N)$. At the base scale,

$$E(w) - \frac{1}{1 + w} = -2w^2 + \mathcal{O}(w^3).$$

Set $w = z^N$ and multiply by $P_m(z) = 1 + \mathcal{O}(z)$. The numerator and denominator degrees are $N - 1$ and N , respectively, so matching through degree $2N - 1$ is exactly the defining Padé condition. $\square$

The poles lie at the roots of -1 , and partial fractions turn the approximant into a finite complex exponential formula for the sequence itself.

Theorem 17.2 (Antiperiodic Gaussian quadrature). *Let*

$$\Lambda_N := \{\lambda \in \mathbb{C} : \lambda^N = -1\}, \quad w_\lambda := \frac{1}{N} P_m(\lambda^{-1}). \quad (159)$$

Then

$$\frac{P_m(z)}{1 + z^N} = \sum_{\lambda \in \Lambda_N} \frac{w_\lambda}{1 - \lambda z}. \quad (160)$$

Consequently, for every integer $0 \leq r < 2N$,

$$\boxed{\varepsilon_r = \frac{1}{N} \sum_{\lambda^N = -1} P_m(\lambda^{-1}) \lambda^r.} \quad (161)$$

The unsigned form is

$$\tau_r = \frac{1}{2} \left[1 - \frac{1}{N} \sum_{\lambda^N = -1} P_m(\lambda^{-1}) \lambda^r \right] \quad (0 \leq r < 2N). \quad (162)$$

Proof. The poles of $P_m(z)/(1 + z^N)$ are $z = \lambda^{-1}$. Multiplying by $1 - \lambda z$ and taking the limit gives the coefficient $P_m(\lambda^{-1})/N$, proving (160). Expand every term as a geometric series. The Padé matching in (157) says that the first $2N$ coefficients equal ε_r , which gives (161). $\square$

Formula (161) is an N -node complex Gaussian quadrature for the formal Thue–Morse moment functional: it reproduces $2N$ consecutive moments. It can also be verified directly from the anti-periodicity $\varepsilon_{N+r} = -\varepsilon_r$ for $0 \leq r < N$ and $\lambda^N = -1$, but the Padé derivation explains why the same nodes and weights are canonical.

The unsigned generating function

$$T(z) = \sum_{n \geq 0} \tau_n z^n = \frac{1}{2} \left(\frac{1}{1 - z} - E(z) \right) \quad (163)$$

has the simultaneous rational approximation

$$\boxed{T(z) = \frac{1 + z^N - (1 - z)P_m(z)}{2(1 - z)(1 + z^N)} + \mathcal{O}(z^{2N}).} \quad (164)$$

17.1 A Krawtchouk and contour avatar

There is a second orthogonal-polynomial representation of the signs, independent of the formal Jacobi system above. Fix any m with $n < 2^m$, and put $r = \text{wt}_2(n)$. The top-degree Krawtchouk coefficient [14] gives

$$\boxed{\varepsilon_n = [u^m](1 - u)^r(1 + u)^{m-r} = {}_2F_1(-m, -r; -m; 2) = (-1)^r.} \quad (165)$$

Equivalently,

$$\varepsilon_n = \frac{1}{2\pi i} \oint_{|u|=\delta} \frac{(1 - u)^r(1 + u)^{m-r}}{u^{m+1}} du, \quad (166)$$

with $0 < \delta < 1$, and

$$\tau_n = \frac{1}{2} (1 - [u^m](1 - u)^r(1 + u)^{m-r}). \quad (167)$$

The coefficient of u^m has only one contributing binomial term, namely the choice of all r minus signs and all $m - r$ plus signs. Although elementary after the fact, this places the signed and unsigned words at an extremal Krawtchouk degree and links their binary digit statistic to a classical hypergeometric family.

18 Numerical experiments and reproducibility

18.1 Exact arithmetic

The script `rvachev_frontier_experiments.py` performs the following operations.

1. It computes $\mu_0, \mu_2, \dots$ exactly using (74).
2. It constructs the monic orthogonal polynomials using the symmetric three-term recurrence and exact rational inner products.
3. It writes all exact β_n through a user-selected order to CSV.
4. It checks (37) independently using a long moment expansion.
5. It compares the cotangent, tangent, and zero-set formulas for Φ'/Φ .
6. It generates Figures 1 and 2.

The default order is 40. On the reference environment the full run takes tens of seconds; the exact integers, rather than floating-point work, dominate at higher order.

18.2 Numerical verification of the resolvent equation

At 45-decimal mpmath precision, using moment expansions through orders 160 and 220, the computed values were:

z	$R'(z)$	$2(R(2z + 1) - R(2z - 1))$	relative discrepancy
1.25	-0.8312850091714830623446214	same	3.6×10^{-46}
1.75	-0.3679189599146538753803797	same	below printed precision
2.50	-0.1691589760955563001740583	same	4.4×10^{-47}

At $z = 1.25$, the change between the two truncation lengths was about 3.8×10^{-41} , so the identity error is below the visible truncation scale.

18.3 Numerical comparison of logarithmic derivatives

The cotangent and tangent formulas agreed to roughly 44 decimal places at the tested nonzero points. The zero-set formula, truncated at 10^5 , differed by 5.2×10^{-6} , 1.5×10^{-5} , and 2.9×10^{-5} at $z = 0.13, 0.37, 0.73$, respectively, consistent with its algebraic tail. The data file `numerical_checks.txt` records all values.

18.4 Second-wave sampling, Mellin, and Padé checks

The companion program `frontier_sampling_pade_experiments.py` independently checks the new formulas at 80-decimal precision and generates all data in `generated/data`. The principal residuals were:

- sparse Shannon and Mittag–Leffler formulas: between 10^{-20} and 10^{-18} with 128 half-lattice coefficients;
- the Bose–Malmsten integral: below 5.1×10^{-15} at real and complex test points;
- the nonlinear formal-Jacobi recursion and $p_{2^m} = x^{2^m} + 1$: exact through degree 96 using rational arithmetic;
- roots-of-minus-one quadrature: below 5×10^{-79} for all moments through level $N = 64$;
- Mellin moment series versus direct cosine-density quadrature: between 10^{-13} and 2.1×10^{-14} for the displayed tests.

No numerical observation is used inside a proof. The exact checks are regression guards against normalization, scaling, and sign errors.

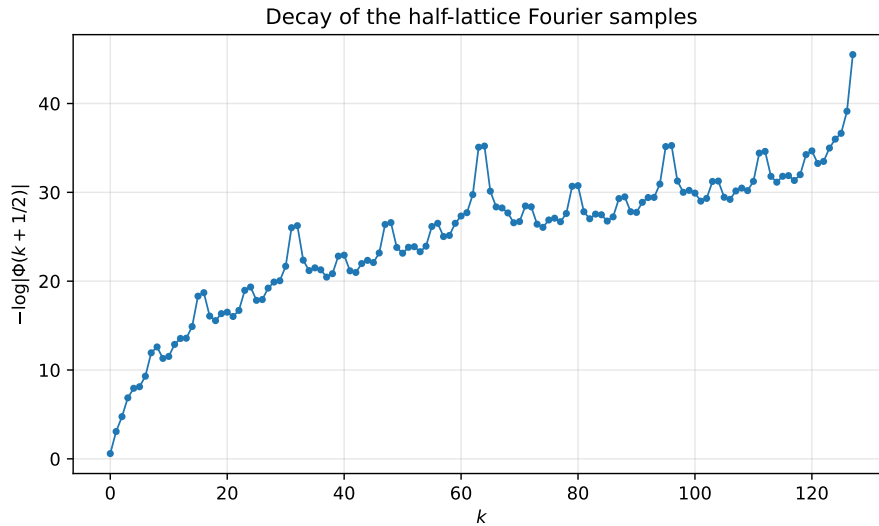


Figure 3: The half-lattice samples decay faster than every power, while their signs are the signed Thue–Morse word. The local irregularity reflects the binary orbit structure.

19 Conjectures and frontier problems

The following conjectures are deliberately separated from the proved results.

Conjecture 19.1 (Strict Jacobi monotonicity). The exact Jacobi coefficients are strictly increasing:

$$0 < \beta_1 < \beta_2 < \cdots < \frac{1}{4}. \quad (168)$$

Exact arithmetic verifies the inequalities through $n = 40$. The upper limit is proved by Theorem 10.4, but monotonicity does not follow from the Denisov–Rakhmanov theorem and appears to require a Fabius-specific argument.

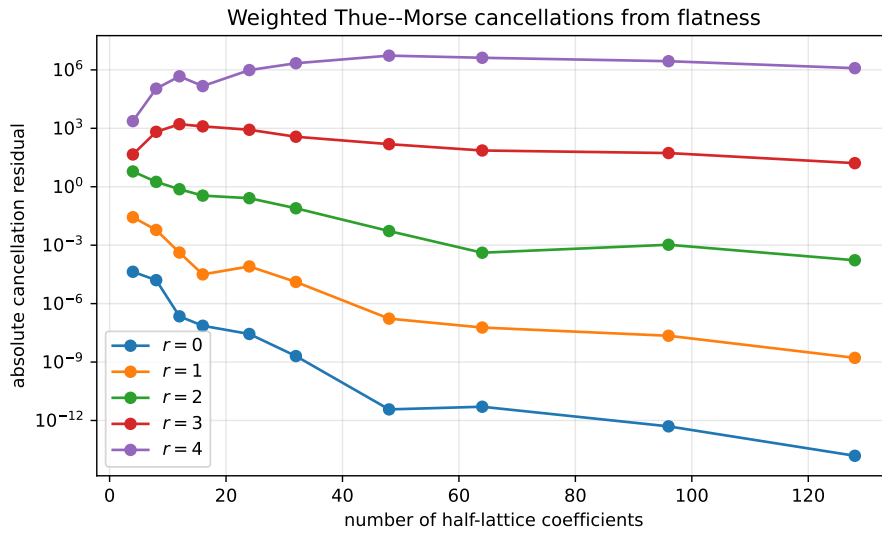


Figure 4: Truncation residuals in the spectral Prouhet identities (116). Higher powers initially amplify the tail before the super-polynomial Fourier decay dominates.

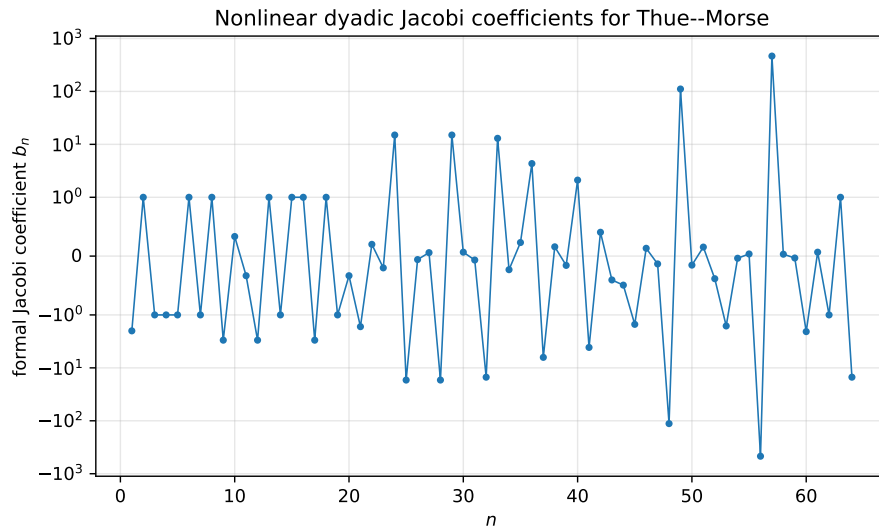


Figure 5: The indefinite formal Jacobi coefficients b_n , shown on a symmetric logarithmic scale. Their nonlinear dyadic recursion creates simultaneous very large and very small rational values.

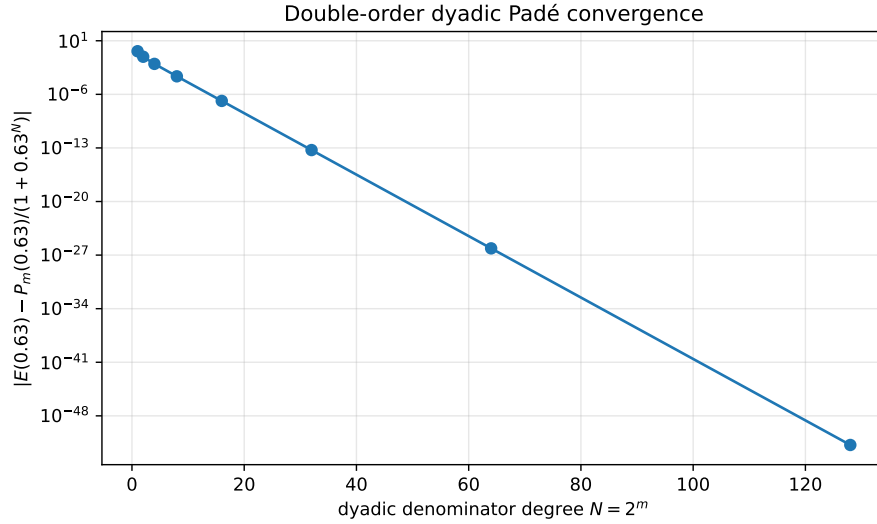


Figure 6: Error of the explicit dyadic Padé approximant at $z = 0.63$. The order doubles with each dyadic level, as proved in Theorem 17.1.

Conjecture 19.2 (Logarithmically corrected recurrence rate). There exists a constant $C_{\text{up}} \in (0, \infty)$ such that

$$\frac{1}{4} - \beta_n \sim C_{\text{up}} \frac{(\log n)^2}{n^2}. \quad (169)$$

The scaled quantity in Figure 2 rises from approximately 0.674 at $n = 20$ to 0.766 at $n = 40$. This is too short a range to determine a constant or even to exclude a different slowly varying factor. The conjecture is motivated jointly by the data and by the squared-logarithmic endpoint scale of ρ . A Riemann–Hilbert or strong Szegő analysis adapted to the endpoint flatness is the natural next step.

Conjecture 19.3 (Dyadically regularized Mittag–Leffler expansion). Let $d_n = 1 + \nu_2(|n|)$, and let $\text{PP}_n(z)$ be the principal part of $1/\Phi(z)$ at the nonzero integer n . There exists a canonical dyadic grouping and a finite genus of subtraction polynomials such that

$$\frac{1}{\Phi(z)} = E(z) + \sum_{n \in \mathbb{Z} \setminus \{0\}}^{\text{dyadic}} (\text{PP}_n(z) - \text{canonical subtraction at } n), \quad (170)$$

where E is an explicitly controlled entire function, ideally constant or of very low order.

The local principal parts are computable from Theorem 8.2; the unresolved issue is a global summation scheme compatible with the unbounded dyadic multiplicities.

Problem 19.4 (Arithmetic of β_n). Determine sharp denominator-clearing factors for β_n , h_n , and the parity Hankel determinants. The first coefficients display repeated migration of prime factors between consecutive norms, suggesting that factored norm variables may be arithmetically simpler than reduced β_n themselves.

Problem 19.5 (Endpoint-sensitive Gaussian nodes). Determine the asymptotic distance of the extreme zero of P_n from 1, and compare it with the Chebyshev scale n^{-2} . The density is positive almost everywhere but flatter than every power at the endpoint, so the leading equilibrium distribution is classical while the first correction should retain the Lambert– W /log-square endpoint signature.

Problem 19.6 (A coupled resolvent hierarchy for exponent sequences). The repository’s Newton/Rvachev theory replaces the unit exponent sequence in the sinc product by general integer sequences. Construct the corresponding Cauchy transforms and derive coupled affine functional equations when the exponent sequence is closed under a Pascal or shift operator. The Fredholm model of Section 9 suggests that the Fourier side is easy; the real-space fixed-point law is the main obstacle.

Problem 19.7 (Fractional resolvent dynamics). Equation (37) gives every integer derivative as a finite Thue–Morse orbit. Seek a fractional-order interpolation

$$D_z^\alpha R(z) = \int_{\Gamma_\alpha} K_\alpha(j) R(2^\alpha z + \text{affine shift}(j)) \, dj$$

that reduces to (38) for $\alpha \in \mathbb{N}$. A semigroup built from the two affine branches $z \mapsto 2z \pm 1$ may provide the correct operator calculus.

Problem 19.8 (Formal verification roadmap). A Lean development can be staged as follows:

1. define the Cauchy transform of the canonical up probability measure;
2. prove the logarithmic fixed-point identity first on $(1, \infty)$;
3. differentiate under the integral and formalize (37);
4. prove the Thue–Morse derivative tree by induction on binary children;
5. define rational moments and the monic Stieltjes procedure;
6. construct finite Jacobi convergents and prove the Gaussian error identity;
7. postpone the Denisov–Rakhmanov limit, which depends on substantial external orthogonal-polynomial theory.

Conjecture 19.9 (Unbounded formal-Jacobi renormalization). For the rational sequence defined by (152),

$$\limsup_{n \rightarrow \infty} |b_n| = \infty, \quad \liminf_{n \rightarrow \infty} |b_n| = 0.$$

Exact computation through several thousand indices shows both effects on every sufficiently large dyadic block. The identity $\prod_{n=2}^{2^m} b_n = 1$, which follows from $h_{2^m} = h_1$, forces a geometric balance but does not by itself prove either extreme.

Conjecture 19.10 (Dyadic weak limit for logarithmic Jacobi coefficients). The empirical measures

$$\nu_m = 2^{-m} \sum_{n=1}^{2^m} \delta_{m^{-2} \log |b_n|}$$

converge weakly to a compactly supported, non-degenerate probability measure. A more refined version should replace the factor m^{-2} by the true height scale and should admit a transfer-operator description induced by (152).

Conjecture 19.11 (No anomalous spectral cancellation). For each fixed $r \geq 0$, the tail in (116) has the same logarithmic order as its largest term along a set of dyadic blocks of density one. Equivalently, apart from exceptional binary rays, the exact global cancellation does not create a second systematic exponential scale beyond the known decay scale of $|\Phi(k + \frac{1}{2})|$.

Problem 19.12 (Formalization package). Formalize Theorems 12.1, 14.1, 16.1, and 17.1 in Lean. The last two require only polynomial algebra, quasi-definiteness imported from finite Hankel determinants, and the binary recursions $\varepsilon_{2n} = \varepsilon_n$, $\varepsilon_{2n+1} = -\varepsilon_n$; they are the most immediate targets. The sampling theorem requires a reusable Paley–Wiener library.

20 Promising research program

20.1 Analytic program

The Cauchy transform should be treated as a primary Fabius–Rvachev object, on the same level as F , ρ , and Φ . Immediate goals are:

1. derive quantitative bounds for R near the cut from the sharp Fourier decay;
2. study the Hilbert transform (35), including its zeros and endpoint asymptotics;
3. use (37) to continue asymptotic expansions along dyadic affine rays;
4. compare finite-logarithmic and Jacobi rational approximants uniformly near the support.

20.2 Orthogonal-polynomial program

The exact J -fraction opens a separate research direction:

1. prove or disprove Conjecture 19.1;
2. determine strong asymptotics of P_n , h_n , and β_n ;
3. construct Fabius-specific Gauss–Kronrod or nested positive rules;
4. investigate whether the dyadic moment recurrence induces a nonlinear recurrence or integrable system for β_n ;
5. study the arithmetic Galois structure of Gaussian nodes and exact weights.

20.3 Operator and special-function program

The Fredholm determinant and Hurwitz-zeta hierarchy suggest:

1. regularized determinants and spectral shifts for deformed multiplicity sequences;
2. a Barnes-type compression of the determinant rather than only of the scalar product;
3. exact trace identities for logarithmic derivatives at rational and half-integer arguments;
4. meromorphic continuation of dyadically weighted spectral sums with local factors dictated by $\nu_2(n)$.

20.4 Inverse-function program

The pushforward formulas (55)–(58) show that many global integral questions about F^{-1} are easier than pointwise questions. Promising directions include:

1. exact orthogonal expansions of $I(u)$ using its known pushforward moments;
2. entropy and transport functionals of I , reduced to integrals against ρ ;
3. combining the global Cauchy transform with local Lambert– W endpoint inversions;
4. rational and Gaussian quadrature for inverse-function integrals without directly evaluating I .

21 Conclusion

The audited repository already contains most of the classical and modern representation theory one would first seek: convolution laws, sinc and cosine products, integer-zero products, moment and cumulant series, Thue–Morse splines, q-binomial dyadic values, Poisson and Legendre expansions, inverse asymptotics, and Fourier-decay envelopes. The most productive way to extend that corpus is therefore to introduce a new organizing transform rather than another isolated formula.

The Cauchy transform R accomplishes this. It joins the moment sequence to an analytic function, converts the dyadic law into a functional-differential equation, turns finite convolutions into logarithmic Thue–Morse sums, and generates a Jacobi continued fraction with exact rational coefficients. The Fourier image simultaneously acquires an operator determinant and a special-function logarithmic-derivative calculus. Closed one-sided transforms and inverse pushforward identities complete a network in which real-space, frequency-space, moment, orthogonal-polynomial, and operator representations are mutually convertible.

The proved limit $\beta_n \rightarrow 1/4$ anchors the new continued-fraction theory. The exact order-40 data, the monotonicity conjecture, the endpoint-sensitive rate problem, and the regularized Mittag–Leffler problem provide concrete frontier targets suitable for both analysis and formalization.

A Reproducibility instructions

From the directory containing this report, run the two independent experiment suites:

```
python3 rvachev_frontier_experiments.py \
  --order 40 --digits 45 --output-dir first_wave_generated

python3 frontier_sampling_pade_experiments.py \
  --output-dir generated
```

The first command regenerates the exact up-law moments, the positive Jacobi coefficients, `generated_tables.tex`, the resolvent/logarithmic-derivative check log, and the two first-wave figures. The second command regenerates the half-lattice samples, weighted Prouhet checks, exact formal-Jacobi coefficients, dyadic Padé/quadrature checks, Mellin and Malmsten checks, and all four second-wave figures under `generated/data` and `generated/figures`.

The exact arithmetic uses Python’s `fractions.Fraction`. The numerical checks use `mpmath` and `scipy`; plotting uses `matplotlib`. Both scripts contain detailed comments, normalization statements, argument validation, and exact regression checks. Packaged root-level copies of the LaTeX fragment and figure PDFs make the report compile without rerunning either program.

B Corpus audit ledger

The pinned 67-path inventory and the additional live draft families used in the audit are listed in the accompanying source fragment `corpus_inventory.tex`.

Pinned 67-source ledger

No.	Path relative to Analysis/FabiusFunction/docs
1	FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2	Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3	Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4	fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5	non-formalized-research-frontiers/non-formalized-research-frontiers.tex
6	archive/early-notes/FabiusAsymptotic/FabiusAsymptotic.tex
7	archive/early-notes/K-foldsummationoverthesignedThue-Morsesequence/ K-foldsummationoverthesignedThue-Morsesequence.tex
8	archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex
9	archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
10	archive/lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex
11	archive/lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
12	archive/primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex
13	archive/primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex
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16	archive/primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
17	archive/primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex
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20	archive/q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex
21	archive/q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex
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25	archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
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28	archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex
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36	archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex
37	archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex
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No.	Path relative to Analysis/FabiusFunction/docs
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57	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/ rvachev_up_fourier_decay-5/Rvachev_Up_Fourier_Decay_Final.tex
58	non-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/ rvachev_up_fourier_decay-6/Rvachev_Up_Fourier_Decay_Final_Synthesis.tex
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61	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
62	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
63	papers/arXiv-1502.06738v2/Lacunary_products.tex
64	papers/arXiv-1702.05442v1/09-Function.tex
65	papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
66	papers/arXiv-2211.13570v1/document.tex
67	papers/ubiq15/ubiq15-corrected.tex

Live additions checked after the pinned ledger

- non-formalized-research-frontiers/drafts/Fabius_Dyadic_Self_Sampling_Frontier_Package/Fabius_Dyadic_Self_Sampling_Frontier_Report.tex
- non-formalized-research-frontiers/drafts/Fabius_Inverse_Frontier_Report_Source_and_PDF/Fabius_Inverse_Frontier_Report.tex
- non-formalized-research-frontiers/drafts/Fabius_Newton_Rvachev_Frontier_Report/Exponent_Sequence_Newton_Rvachev_Frontiers.tex
- non-formalized-research-frontiers/drafts/fabius_frontier_dyadic_inverse_barnes_report/ (report source and auxiliaries)

Additional auxiliary TeX fragments inside these live report packages were checked through their parent report structure. The pinned ledger remains the stable reproducible count; the live tree changes during active research and therefore should not be assigned a permanent count without a fresh commit-pinned traversal.

C Selected exact data

The first eleven even moments and first twelve Jacobi coefficients are typeset in the main text. The CSV files contain the exact data through order 40. Selected late-order decimal values are:

n	β_n	$1/4 - \beta_n$	$n^2(1/4 - \beta_n)/(\log n)^2$
12	0.223692820061698	0.026307179938302	0.6135029154
15	0.229185734718218	0.020814265281782	0.6386018638
20	0.234873012373992	0.015126987626008	0.6742274757
25	0.238343193605764	0.011656806394236	0.7031551379
30	0.240650640426945	0.009349359573055	0.7273789024
35	0.242285352613889	0.007714647386111	0.7476320691
40	0.243489365492140	0.006510634507860	0.7655161747

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